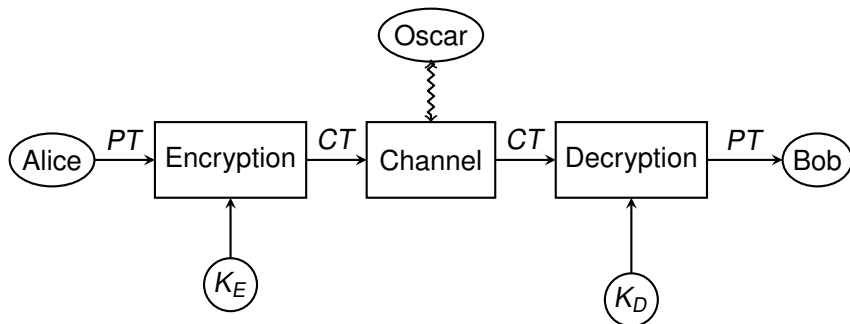


Construction of Boolean functions and S-boxes with evolutionary algorithms

Courtesy of Luca Mariot, Radboud University – Twente University

Cryptography

Basic Goal of Cryptography: Enable two parties (Alice and Bob, A and B) to securely communicate over an insecure channel, even in presence of an opponent (Oscar, O)



▶ PT: plaintext

▶ CT: ciphertext

▶ K_E : encryption key

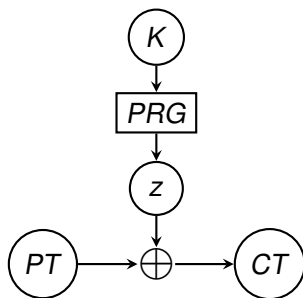
▶ K_D : decryption key

Symmetric cryptosystems

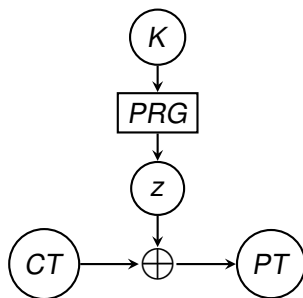
Symmetric cryptosystems ($K_E = K_D = K$) can be classified as:

- ▶ *Stream ciphers*: each symbol of PT is combined with a symbol of a *keystream*, computed from K
 - ▶ GRAIN
 - ▶ TRIVIUM
 - ▶ ...
- ▶ *Block ciphers*: PT is divided in *blocks* combined with *round keys* derived from K through a *round function*
 - ▶ DES
 - ▶ RIJNDAEL (AES)
 - ▶ ...

Vernam Stream Cipher



(a) Encryption



(b) Decryption

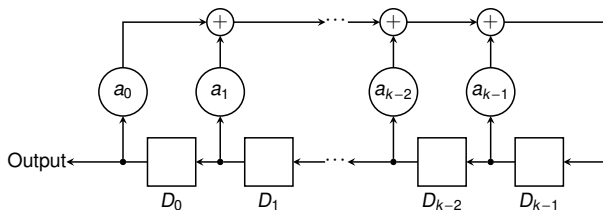
- ▶ K : secret key
- ▶ PRG: Pseudorandom Generator
- ▶ z : keystream

- ▶ $\oplus$: bitwise XOR
- ▶ PT : Plaintext
- ▶ CT : Ciphertext

Linear Feedback Shift Registers (LFSR)

- ▶ Device computing the **binary linear recurring sequence**

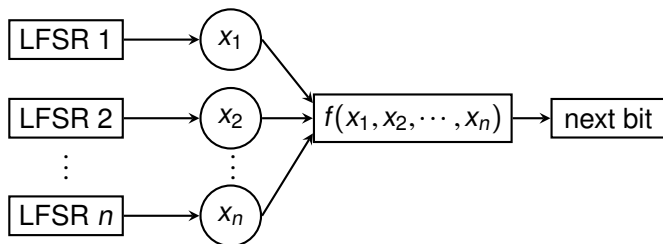
$$s_{n+k} = a + a_0 s_n + a_1 s_{n+1} + \dots + a_{k-1} s_{n+k-1}$$



- ▶ **Too weak** as a PRG: $2k$ consecutive bits of keystream are enough to recover the LFSR initialization via the **Berlekamp-Massey algorithm**

An Example of PRG: The Combiner Model

- ▶ a **Boolean function** $f : \{0, 1\}^n \rightarrow \{0, 1\}$ combines the outputs of n LFSR [1]



- ▶ Security of the combiner $\Leftrightarrow$ **cryptographic properties** of f

Boolean Functions - Basic Definitions

Boolean function: a mapping $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$, where $\mathbb{F}_2 = \{0, 1\}$

- ▶ **Truth table:** vector Ω_f specifying $f(x)$ for all $x \in \mathbb{F}_2^n$

(x_1, x_2, x_3)	000	100	010	110	001	101	011	111
Ω_f	0	1	1	1	1	0	0	0

- ▶ **Algebraic Normal Form (ANF):** Sum (XOR) of products (AND) over the finite field $\mathbb{F}_2$

$$f(x_1, x_2, x_3) = x_1 \cdot x_2 \oplus x_1 \oplus x_2 \oplus x_3$$

- ▶ **Walsh Transform:** correlation with the *linear* functions defined as $\omega \cdot x = \omega_1 x_1 \oplus \dots \oplus \omega_n x_n$

$$\hat{F}(\omega) = \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x) \oplus \omega \cdot x}$$

Cryptographic Properties: Balancedness

- ▶ **Hamming weight** $w_H(f)$: number of 1s in Ω_f
- ▶ A function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ is **balanced** if $w_H(f) = 2^{n-1}$
- ▶ Walsh characterization: f balanced $\Leftrightarrow \hat{F}(0) = 0$

(x_1, x_2, x_3)	000	100	010	110	001	101	011	111
Ω_f	0	1	1	1	1	0	0	0



f is balanced

- ▶ Unbalanced functions present a statistical bias that can be exploited in attacks

Cryptographic Properties: Algebraic Degree

- ▶ **Algebraic degree** d : the degree of the multivariate polynomial representing the ANF of f

$$f(x_1, x_2, x_3) = x_1 \cdot x_2 \oplus x_1 \oplus x_2 \oplus x_3$$



f has degree $d = 2$

- ▶ *Linear* functions $\omega \cdot x = \omega_1 x_1 \oplus \dots \oplus \omega_n x_n$ have degree $d = 1$
- ▶ Boolean functions of high degree make the attack based on Berlekamp-Massey algorithm less effective

Cryptographic Properties: Nonlinearity

- ▶ **Nonlinearity** $nl(f)$: Hamming distance of f from linear functions
- ▶ Walsh characterization:

$$nl(f) = 2^{n-1} - \frac{1}{2} \max_{\omega \in \mathbb{F}_2^n} \{|\hat{F}(\omega)|\}$$

(x_1, x_2, x_3)	000	100	010	110	001	101	011	111
Ω_f	0	1	1	1	1	0	0	0
$\hat{F}(\omega)$	0	0	0	0	-4	4	4	4

⇓

$$nl(f) = 2^{3-1} - \frac{1}{2} \cdot 4 = 2$$

- ▶ Functions with high nonlinearity resist **fast-correlation attacks**

Cryptographic Properties: Resiliency

- ▶ ***t*-Resiliency**: when fixing any t variables, the restriction of f stays balanced
- ▶ Walsh characterization:

$$\hat{F}(\omega) = 0 \quad \forall \omega : w_H(\omega) \leq t$$

(x_1, x_2, x_3)	000	100	010	110	001	101	011	111
Ω_f	0	1	1	1	1	0	0	0
$\hat{F}(\omega)$	0	0	0	0	-4	4	4	4



$$F(001) = -4 \Rightarrow f \text{ is NOT 1-resilient}$$

- ▶ Resilient functions of high order t resist to **correlation attacks**

Bounds and Trade-offs

In summary, $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ should:

- ▶ be balanced
- ▶ be resilient of high order m
- ▶ have high algebraic degree d
- ▶ have high nonlinearity nl

But most of these properties cannot be satisfied simultaneously!

- ▶ *Covering Radius bound*: $nl \leq 2^{n-1} - 2^{\frac{n}{2}-1}$
- ▶ *Siegenthaler's bound*: $d \leq n - t - 1$
- ▶ *Tarannikov's bound*: $nl \leq 2^{n-1} - 2^{t+1}$

Constructions of good Boolean Functions

- ▶ Number of Boolean functions of n variables: 2^{2^n}
- ▶ $\Rightarrow$ too huge for exhaustive search when $n > 5$!
- ▶ Functions used in the combiner model have $n \geq 13$ variables

In practice, one usually resorts to:

- ▶ Algebraic constructions [1]
 - ▶ *Maierana-McFarland construction*
 - ▶ *Rothaus' construction*
 - ▶ ...
- ▶ Heuristic techniques
 - ▶ *Simulated Annealing* [3]
 - ▶ *Evolutionary Algorithms* [5]
 - ▶ ...

Evolutionary Search of Boolean Functions

- ▶ **Evolutionary search** offers a promising way to optimize cryptographic boolean functions
- ▶ Usual approach: directly search the space of truth tables, represented as 2^n -bit strings [6]
- ▶ Fitness function measuring nonlinearity, algebraic degree, and deviation from correlation-immunity
- ▶ Specialized variation operators for preserving balancedness

- ▶ Applying the Inverse Walsh Transform to a generic spectrum yields a **pseudoboolean function** $f : \mathbb{F}_2^n \rightarrow \mathbb{R}$

$$\mathcal{S}_f = (0, -4, -2, 2, 2, 4, 4, -2)$$

$$\Downarrow \hat{F}^{-1}$$

$$\Omega_{\hat{f}} = (0, 0, 0, -1, 0, -1, 2)$$

- ▶ **New objective**: minimize the **deviation** of Walsh spectra which satisfy the desired cryptographic constraints
- ▶ Heuristic techniques proposed for this optimization problem:
 - ▶ Clark et al. [2]: Simulated Annealing (SA)
 - ▶ Mariot and Leporati [5]: Genetic Algorithms (GA)

Plateaued Functions [8]

- ▶ Our GA evolves spectra of **plateaued** functions
- ▶ A (pseudo)boolean function f is plateaued if its Walsh spectrum takes only three values: $-W_M(f)$, 0 and $+W_M(f)$

$$S_f = (0, 0, 0, 0, -4, 4, 4, 4) \Rightarrow \text{plateaued}$$

- ▶ Motivations:
 - ▶ Simple combinatorial representation of candidate solutions, determined by a single parameter $r \geq n/2$
 - ▶ Plateaued functions reach both Siegenthaler's and Tarannikov's bounds

Chromosome Encoding

- ▶ **Resiliency Constraint:** ignore positions with at most m ones

x	<u>000</u>	<u>100</u>	<u>010</u>	110	<u>001</u>	101	011	111
S_f	0	0	0	-4	0	4	4	4

- ▶ The **chromosome** c is the permutation of the spectrum in the positions with more than m ones:

x	110	101	011	111
c	-4	4	4	4

- ▶ The multiplicities of 0, $-W_M(f)$ and $+W_M(f)$ in the permutation depend on plateau index r

Fitness Function

- ▶ Given $\hat{f} : \mathbb{F}_2^n \rightarrow \mathbb{R}$, the **nearest boolean function** $\hat{b} : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ is defined for all $x \in \mathbb{F}_2^n$ as:

$$\hat{b}(x) = \begin{cases} +1 & , \text{ if } \hat{f}(x) > 0 \\ -1 & , \text{ if } \hat{f}(x) < 0 \\ +1 \text{ or } -1 \text{ (chosen randomly)} & , \text{ if } \hat{f}(x) = 0 \end{cases}$$

- ▶ **Objective function** proposed in [2]:

$$obj(f) = \sum_{x \in \mathbb{F}_2^n} (\hat{f}(x) - \hat{b}(x))^2$$

- ▶ **Fitness function** maximised by our GA: $fit(f) = -obj(f)$

- ▶ **Crossover** between two Walsh spectra p_1, p_2 must preserve the multiplicities of $-W_M(f)$, 0 and $+W_M(f)$
- ▶ **Idea**: use counters to keep track of the multiplicities [6]
- ▶ **Mutation**: swap two random positions in the chromosome with **different** values
- ▶ **Selection** operators adopted:
 - ▶ **Roulette-Wheel** (*RWS*)
 - ▶ **Deterministic Tournament** (*DTS*)

Experimental Settings

Common parameters:

- ▶ Number of variables $n = 6, 7$ and plateau index $r = 4$

(n, m, d, nl)	$ 0_{res} $	$ 0_{add} $	$ -W_M(f) $	$ +W_M(f) $
(6, 2, 3, 24)	22	26	6	10
(7, 2, 4, 56)	29	35	28	36

GA-related parameters:

- ▶ Population size $N = 30$
- ▶ max generations $G = 500000$
- ▶ GA runs $R = 500$
- ▶ Crossover probability $p_\chi = 0.95$
- ▶ Mutation probability $p_\mu = 0.05$
- ▶ Tournament size $k = 3$

SA-related parameters:

- ▶ Inner loops $MaxIL = 3000$
- ▶ Moves in loop $MIL = 5000$
- ▶ SA runs $R = 500$
- ▶ Initial temperatures $T = 100, 1000$
- ▶ Cooling parameter: $\alpha = 0.95, 0.99$

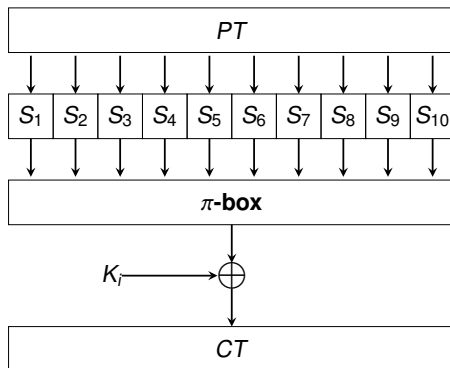
Results

Statistics of the best solutions found by our GA and SA over $R = 500$ runs.

n	Stat	GA(RWS)	GA(DTS)	SA(T_1, α_1)	SA(T_2, α_2)
6	avg_o	14.08	13.02	19.01	19.03
	min_o	0	0	0	0
	max_o	16	16	28	28
	std_o	5.21	6.23	4.89	4.81
	$\#opt$	60	93	11	10
	avg_t	83.3	79.2	79.1	79.4
7	avg_o	53.44	52.6	45.09	44.85
	min_o	47	44	32	27
	max_o	58	59	63	57
	std_o	2.40	2.77	4.39	4.18
	$\#opt$	0	0	0	0
	avg_t	204.2	204.5	180.3	180.2

Block Ciphers: Substitution-Permutation Network

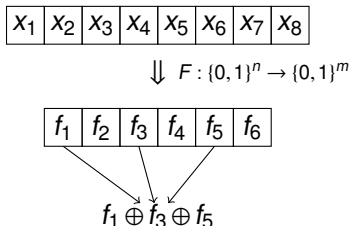
Round function of a SPN cipher:



- ▶ $S_i : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ are **S-boxes** providing **confusion** [7]
- ▶ Security of confusion layer $\Leftrightarrow$ cryptographic properties of S_i

Background on S-boxes (1/2)

- ▶ A **Substitution Box** (S-box) is a mapping $F : \{0, 1\}^n \rightarrow \{0, 1\}^m$ defined by m **coordinate functions** $f_i : \{0, 1\}^n \rightarrow \{0, 1\}$
- ▶ The **component functions** $v \cdot F : \{0, 1\}^n \rightarrow \{0, 1\}$ for $v \in \{0, 1\}^m$ of F are the **linear combinations** of the f_i



- ▶ The **nonlinearity** of a S-box F is defined as the minimum nonlinearity among all its component functions
- ▶ S-boxes with high nonlinearity allow to resist to **linear cryptanalysis** attacks

Background on S-Boxes (2/2)

- ▶ **delta difference table** of F wrt a, b :

$$D_F(a, b) = \{x \in \mathbb{F}_2^n : F(x) \oplus F(x \oplus a) = b\}.$$

- ▶ Given $\delta_F(a, b) = |D_F(a, b)|$, the **differential uniformity** of F is:

$$\delta_F = \max_{\substack{a \in \{0, 1\}^{n*} \\ b \in \{0, 1\}^m}} \delta_F(a, b).$$

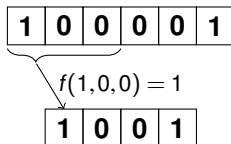
- ▶ S-boxes with low differential uniformity are able to resist **differential cryptanalysis attacks**

Cellular Automata S-boxes

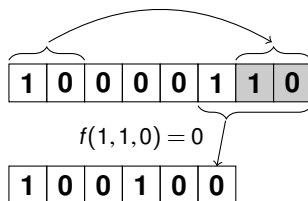
- ▶ One-dimensional **Cellular Automaton** (CA): a discrete parallel computation model composed of a finite array of n **cells**
- ▶ Each cell updates its **state** $s \in \{0, 1\}$ by applying a **local rule** $f : \{0, 1\}^d \rightarrow \{0, 1\}$ to itself and the $d - 1$ cells to its right

Example: $n = 6$, $d = 3$, $f(s_i, s_{i+1}, s_{i+2}) = s_i \oplus s_{i+1} \oplus s_{i+2}$,

Truth table: $\Omega(f) = 01101001 \rightarrow \text{Rule 150}$



No Boundary CA – NBCA



Periodic Boundary CA – PBCA

Problem Statement

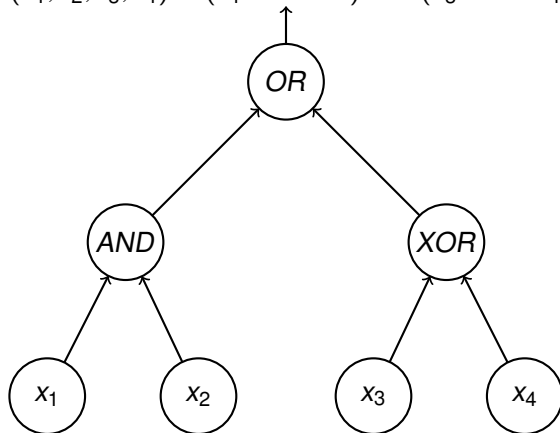
- ▶ **Goal:** Find PBCA of length n and diameter $d = n$ having cryptographic properties equal to or better than those of other real-world S-boxes
- ▶ Considered S-boxes sizes: from $n = 4$ to $n = 8$
- ▶ Using **tree encoding**, exhaustive search is already unfeasible for $n = 4$
- ▶ We adopted **Genetic Programming** to address this problem

Genetic Programming (GP)

- ▶ Optimization method inspired by evolutionary principles, introduced by Koza [4]
- ▶ Each candidate solution (individual) is represented by a **tree**
 - ▶ Terminal nodes: input variables
 - ▶ Internal nodes: Boolean operators (AND, OR, NOT, XOR, ...)
- ▶ New solutions are created through genetic operators like **tree crossover** and **subtree mutation** applied to a population of candidate solutions
- ▶ Optimization is performed by evaluating the new candidate solutions wrt a **fitness function**

GP Tree Encoding – Example

$$f(x_1, x_2, x_3, x_4) = (x_1 \text{ AND } x_2) \text{ OR } (x_3 \text{ XOR } x_4)$$



- ▶ Considered cryptographic properties:
 - ▶ balancedness/invertibility ($BAL = 0$ if F is balanced, -1 otherwise)
 - ▶ nonlinearity N_F
 - ▶ differential uniformity δ_F
- ▶ **Fitness function** maximized:

$$fitness = BAL + \Delta_{BAL,0} \left(N_F + \left(1 - \frac{nMinN_F}{2^n} \right) + (2^n - \delta_F) \right).$$

where $\Delta_{BAL,0} = 1$ if F is balanced and 0 otherwise, and $nMinN_F$ is the number of occurrences of the current value of nonlinearity

Experimental Setup

- ▶ Problem instance / CA size: $n = 4$ up to $n = 8$
- ▶ Maximum tree depth: equal to n
- ▶ Genetic operators: simple tree crossover, subtree mutation
- ▶ Population size: 2000
- ▶ Stopping criterion: 2000000 fitness evaluations
- ▶ Parameters determined by initial tuning phase on $n = 6$ case

Table: Statistical results and comparison.

S-box size	T_{max}	GP			N_F	δ_F
		Max	Avg	Std dev		
4×4	16	16	16	0	4	4
5×5	42	42	41.73	1.01	12	2
6×6	86	84	80.47	4.72	24	4
7×7	182	182	155.07	8.86	56	2
8×8	364	318	281.87	13.86	82	20

- ▶ From $n = 4$ to $n = 7$, we obtained CA rules inducing S-boxes with optimal crypto properties
- ▶ Only for $n = 8$ the performances of GP are consistently worse wrt to the theoretical optimum

- ▶ Boolean functions and S-boxes play a fundamental role in the design of symmetric ciphers
- ▶ The design of Boolean functions and S-boxes with good properties is a hard optimization problem
- ▶ For Boolean functions, GA are more efficient than SA under the spectral inversion approach
- ▶ For S-boxes, GP is able to find optimal solutions up to size 7×7

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